FEATURE ENGINEERING, MODEL OPTIMIZATION & PERFORMANCE COMPARISON

House Price Prediction using California Housing Dataset

ABSTRACT

This project focuses on implementing feature engineering, preprocessing, model optimization, and performance comparison techniques for predicting house prices using the California Housing Dataset. The workflow includes data loading, preprocessing, feature scaling, model training, prediction, evaluation, and visualization. Multiple regression algorithms such as Linear Regression, Ridge Regression, and Decision Tree Regressor were implemented and compared.

Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn were used for implementation. The models were evaluated using Root Mean Squared Error (RMSE) and R^2 Score. The results showed that the Decision Tree Regressor achieved better prediction performance compared to Linear Regression and Ridge Regression. This project demonstrates the importance of preprocessing and model comparison in improving machine learning performance.

1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and make predictions automatically. Regression algorithms are supervised learning techniques used for predicting continuous numerical values.

Feature engineering and preprocessing are important stages in machine learning because raw datasets often contain features with different numerical ranges and distributions. Proper preprocessing improves model accuracy and learning efficiency.

In this project, multiple regression models were developed to predict house prices using the California Housing Dataset. The dataset contains information related to housing districts such as median income, house age, average rooms, population, occupancy, latitude, and longitude.

The project demonstrates a complete machine learning workflow including:

- Data preprocessing
- Feature scaling
- Model training
- Prediction
- Evaluation
- Performance comparison
- Visualization

2. DATA PREPROCESSING AND FEATURE ENGINEERING

Data preprocessing and feature engineering techniques were applied before model training.

The following preprocessing steps were carried out:

- Loading the California Housing Dataset
- Separating features and target variable
- Feature scaling using StandardScaler
- Train-test splitting

Observations

- The dataset contained numerical housing-related attributes.
- Feature scaling improved learning stability.
- Standardization helped normalize feature ranges.
- Preprocessed data improved model performance.

3. MODEL BUILDING

The California Housing Dataset was divided into:

- Features (X)
- Target Variable (y)

The dataset was split into training and testing sets using an 80:20 ratio.

Three regression models were trained and compared:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Linear Regression

Linear Regression establishes a linear relationship between independent variables and the target variable.

Ridge Regression

Ridge Regression applies regularization to reduce overfitting and improve generalization.

Decision Tree Regressor

Decision Tree Regressor captures non-linear relationships using tree-based decision rules.

After training, predictions were generated using the testing dataset.

The following Python libraries were used:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn

4. EVALUATION METRICS

The models were evaluated using the following regression metrics:

- Root Mean Squared Error (RMSE)
- R^2 Score

Model Performance Comparison

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

Figure 3: Actual vs Predicted House Prices

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Model Comparison Results
                        RMSE   R2 Score
Linear Regression   0.745581   0.575788
Ridge Regression    0.745554   0.575819
Decision Tree       0.724234   0.599732

Best Performing Model: Decision Tree
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Figure 4: Model Performance Comparison Output

Interpretation

- Linear Regression provided baseline prediction performance.
- Ridge Regression improved model stability using regularization.

- Decision Tree Regressor achieved lower RMSE and higher R^2 Score.
- The Decision Tree model captured complex relationships more effectively.

The Decision Tree Regressor performed better compared to the other regression models.

5. IMPROVEMENT IDEAS

The performance of the models can be improved further using:

- Hyperparameter tuning
- Cross-validation
- Feature selection
- Outlier removal
- Advanced feature engineering

Advanced machine learning algorithms such as:

- Random Forest Regression
- Gradient Boosting
- XGBoost
- Support Vector Regression

can also be implemented to improve prediction accuracy.

6. CONCLUSION

This project successfully demonstrated feature engineering, preprocessing, model optimization, and performance comparison using the California Housing Dataset.

The workflow included:

- Data loading
- Feature scaling
- Data preprocessing
- Model training
- Prediction
- Evaluation
- Visualization
- Performance comparison

The results showed that preprocessing and model comparison play an important role in improving machine learning performance. Among all the models, the Decision Tree Regressor achieved the best performance with lower RMSE and higher R^2 Score.

The project also provided practical understanding of regression analysis, feature engineering, and machine learning workflows using Python and Scikit-learn.